



FUNCTIONAL DESIGN SPECIFICATION

Sun Metals Z20 — Robotic Ingot- Stacking Cell

Robotic Ingot-Stacking Cell — reverse-engineered from Studio 5000 (KUKA_ROBOT program)

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Prepared by	Pac Technologies Pty Ltd
Control platform	Allen-Bradley CompactLogix 5069-L310ERM

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SECTION 1

1 Document control

1.1 Revision history

Rev	Date	Author	Description
01	20 July 2026	Pac Technologies	First issue — reverse-engineered as-is baseline

SECTION 2

2 Introduction & scope

2.2 Design principles

- PackML-style unit state model (custom BASE_CONTROL on the TOP tag).
- Reusable device AOIs (FB_DI_Sensor / FB_AI_Sensor / FB_DI_PushButton / FB_D_Solenoid) as control modules.
- PGNO program-number handshake to the KUKA robot for all motion.
- Reverse-engineered from L5K; robot motion detail resides in the KUKA .src.

2.3 Exclusions

- Strapping_Machine / OLD_Strapping_Machine / Channel1Program (other units on the same controller).
- Slab_1_M580 CIP bridge (not referenced by KUKA_ROBOT program).
- KUKA .src robot motion programs (referenced by PGNO only).
- Functional safety design (safety controller / GuardLogix).

SECTION 3

3 System architecture

3.1 Network devices

Device	Catalog / type	Address	Role
Balluff IO-Link Master 1 (K_RIO1)	io_link_master	192.168.1.21	Balluff BNI006A EtherNet/IP IO-Link master — 8 ports: P1–P4 IO-Link (laser + DI), P5–P8 DI.
Balluff IO-Link Master 2 (K_RIO2)	io_link_master	192.168.1.22	Balluff BNI006A EtherNet/IP IO-Link master — 8 ports DI (sensors 13–22).
KUKA KRC5 Robot	robot_controller	192.168.1.24	KUKA KRC5 robot controller — PGNO task dispatch, drives/enable handshake, gripper ops, stack/makeup reflections.
SICK Outfeed Encoder	encoder	192.168.1.25	SICK AFM60A EtherNet/IP absolute encoder — tracks stack X-position during indexing (read via explicit MSG).
SMC Valve Manifold	valve_manifold	192.168.1.23	SMC EX260-SEN1 EtherNet/IP valve manifold — 5 solenoid points: grippers A–D + air-blast.

SECTION 4

4 Control philosophy & modes of operation

4.1 Operating modes

Mode	Kind	Description
Automatic default	production	Normal production — full authored state model.
Manual	manual	Operator-paced jogging of grippers/robot via HMI.
Maintenance	maintenance	Robot to maintenance position (Task 7); drives commanded to stopped.

4.2 Fault philosophy

14-word fault table (TOP.CTRL.Fault[0..13]). E-Stop (F0.30/F0.31) = MajorFault forcing all EMs to aborted; other conditions latch as minor faults until Fault_Reset_Request.

4.3 Safety

E-Stop major-fault path (site safety controller scope not in this program)

SECTION 5

5 I/O schedule

Tag	Type	Signal	Address	Description
ESTOP_1	pushbutton	DI	K_RI01:I.Data.x	E-Stop 1
PB_START	pushbutton	DI	K_RI01:I.Data.x	Start pushbutton
PB_STOP	pushbutton	DI	K_RI01:I.Data.x	Stop pushbutton
PB_PAUSE	pushbutton	DI	K_RI01:I.Data.x	Pause pushbutton
INFEED_TopSensor	photoelectric	DI	K_RI02:I.Data.8	Infeed top sensor
INFEED_LowerSensor	photoelectric	DI	K_RI02:I.Data.9	Infeed lower sensor
LASER_A	laser_distance	AI	K_RI01:P1 (I0-Link)	Position 1 laser height scanner
LASER_B	laser_distance	AI	K_RI01:P2 (I0-Link)	Position 2 laser height scanner
LASER_C	laser_distance	AI	K_RI01:P3 (I0-Link)	Position 3 laser height scanner
LASER_D	laser_distance	AI	K_RI01:P4 (I0-Link)	Position 4 laser height scanner
GRIP_A_EXT	solenoid	DO	K_VALVE_MANIFOLD.Point[4].B	Gripper A extend
LASER_AIRBLAST	solenoid	DO	K_VALVE_MANIFOLD.Point[3].A	Laser air- blast

SECTION 6

6 Functional descriptions

6.1 Infeed Queue & Detection

Detects ingots at 4 conveyor pickup positions, classifies Foot/Standard/Top/Flip, maintains the FIFO infeed queue and tracking array, deletes picked ingots and drives the turn-over flipper. Source: R201/R202/R203 + FB_INGOT_DETECTION, FB_IngotDetect_Scan.

Control module	Class	Description
Infeed Top Sensor	digital_input_sensor	Infeed top-level ingot presence (FB_DI_SensorData[21]).
Infeed Lower Sensor	digital_input_sensor	Infeed lower-level ingot presence (FB_DI_SensorData[22]).
Position 1 Top Sensor	digital_input_sensor	Pickup position 1 top-detection (FB_DI_SensorData[13]).
Position 1 Foot Sensor	digital_input_sensor	Pickup position 1 foot-detection (FB_DI_SensorData[17]).
Balluff IO-Link Master 1 (K_RIO1)	io_link_master	Balluff BNI006A EtherNet/IP IO-Link master — 8 ports: P1-P4 IO-Link (laser + DI), P5-P8 DI.
Balluff IO-Link Master 2 (K_RIO2)	io_link_master	Balluff BNI006A EtherNet/IP IO-Link master — 8 ports DI (sensors 13-22).

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: RunSYS = true

Step	Action	Completion criteria
1	Read per-position top/foot sensors + laser scan; FB_INGOT_DETECTION classifies each position as Foot(1)/Standard(2)/Top/Flip(3).	Ingot type resolved for each active position

Step	Action	Completion criteria
2	Insert classified ingot type into INFEED_Queue at the insert index; append to tracking array; raise INFEED_Alarm_Ingot_unexpected if type != expected.	Ingot appended to queue
3	When a stack section signals ACTIONS.7 (picked), delete taken ingots from the queue and compact (R202 FOR loop).	Queue compacted after pick

Turn-over flipper path (INFEED_KUKA_Turn_Over_Flipper, triggered after 36 tracked / every 4 tops) is heuristic — confirm intended rule.

6.2 Robot Interface & Task Executor

Maps the full KUKA KRC5 EtherNet/IP handshake, selects and dispatches the robot task via the PGNO program-number interface, enforces ready/interlock conditions and reflects robot status to PLC/HMI.

Source: R004/R204 + FB_EXECUTE_R_STEPS_ACTIONS, AOI_KRC5_KUKA.

Control module	Class	Description
KUKA KRC5 Robot	robot_controller	KUKA KRC5 robot controller — PGNO task dispatch, drives/enable handshake, gripper ops, stack/makeup reflections.

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: TOP.CTRL.Automatic_Run = true AND KUKA_ROBOT.In.PERI_RDY = true AND KUKA_ROBOT.In.STOPMESS = false

Step	Action	Completion criteria
1	Latch the requested task number (1..8) into Task_Request from the active stack section / HMI IN_TASK_n_EXE.	Task_Request != 0
2	On KUKA_ROBOT.In.PGNO_REQ, assert PGNO_VALID; robot acknowledges and starts the program.	Robot PRO_ACT && APPL_RUN

Step	Action	Completion criteria
3	Robot runs task (move/pick/place/dump). Wait for task-completed handshake (PGNO_REQ low & APPL_RUN).	Task completed
4	Clear Task_Request; update Last_Run_Task; hand back to the calling stack section.	Task_Request cleared

Task map (PGNO): 1=move-to-pick, 2=stack-with-makeup, 3/4=stack variants, 5=send-data, 6=conveytech-release, 7=maintenance. Tasks 3/4 motion detail lives in the KUKA .src — [ASSUMPTION].

6.3 Stack Builder — Foot Ingots

Step/action sequence placing foot ingots onto the active stack; assigns active stack, dispatches robot tasks, advances CurrentLayer/NextLayer. Source: FOOT_SAS/FOOT_SECTION step-action FB.

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: TOP.CTRL.Automatic_Run = true AND FOOT_SAS.QUEUE_FULL = true

Step	Action	Completion criteria
1	Assert ACTIONS.2 → robot Task 1 (move to position); activate grippers per LOCn_PICK.	Robot picked ingots from conveyor
2	Robot places ingots on the active stack at CurrentLayer; ACTIONS.4 sends stack data to robot.	Layer placed
3	ACTIONS.7 marks layer complete, deletes picked ingots from the infeed queue, advances NextLayer; rotate active stack when CurrentLayer >= 10.	Layer count advanced

Foot layer is the bottom layer; ACTIONS map: 2=pick,4=send-data,7=update,11=Task2,20=Task3,26=Task4.

6.4 Stack Builder — Regular Ingots

Step/action sequence placing regular ingots onto the active stack; assigns active stack, dispatches robot tasks, advances CurrentLayer/NextLayer. Source: REGULAR_SAS/REGULAR_SECTION step-

action FB.

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: TOP.CTRLAutomatic_Run = true AND REGULAR_SAS.QUEUE_FULL = true

Step	Action	Completion criteria
1	Assert ACTIONS.2 → robot Task 1 (move to position); activate grippers per LOCn_PICK.	Robot picked ingots from conveyor
2	Robot places ingots on the active stack at CurrentLayer; ACTIONS.4 sends stack data to robot.	Layer placed
3	ACTIONS.7 marks layer complete, deletes picked ingots from the infeed queue, advances NextLayer; rotate active stack when CurrentLayer >= 10.	Layer count advanced

Regular (mid) layers between foot and top.

6.5 Stack Builder — Top Ingots

Step/action sequence placing top ingots onto the active stack; assigns active stack, dispatches robot tasks, advances CurrentLayer/NextLayer. Source: TOP_SAS/TOP_SECTION step-action FB.

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: TOP.CTRLAutomatic_Run = true AND TOP_SAS.QUEUE_FULL = true

Step	Action	Completion criteria
1	Assert ACTIONS.2 → robot Task 1 (move to position); activate grippers per LOCn_PICK.	Robot picked ingots from conveyor
2	Robot places ingots on the active stack at CurrentLayer; ACTIONS.4 sends stack data to robot.	Layer placed
3	ACTIONS.7 marks layer complete, deletes picked ingots from the infeed queue, advances NextLayer; rotate active stack when CurrentLayer >= 10.	Layer count advanced

Top layer uses flip-detected ingots; gated by CurrentLayer==9 & laser slot==1.

6.6 Stack Management & Outfeed Index

Per-stack layer/quantity counting, active-stack rotation, stack-off-conveyor shift to the strapper, and X-position tracking via the SICK encoder. Source: R301 + STACK_COUNT, MAKEUP_COUNT, FB_OUTFEED_ENC_TRACK.

Control module	Class	Description
SICK Outfeed Encoder	encoder	SICK AFM60A EtherNet/IP absolute encoder — tracks stack X-position during indexing (read via explicit MSG).

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: TOP.CTRL.Run = true

Step	Action	Completion criteria
1	STACK_COUNT per stack maintains CurrentLayer/counts; mark Complete when CurrentLayer >= 10.	Stack marked complete
2	On StackOffConeyor, shift STACK[2]→[1], [3]→[2] (backups forward).	Stacks rotated

Step	Action	Completion criteria
3	FB_OUTFEED_ENC_TRACK integrates encoder delta into STACK[n].X_Position_Act during index.	Index distance reached

6.7 Makeup Table

Partial-layer buffer holding top/mid/btm racks with required quantity per option; fills when no stack is active. Source: MAKEUP_TABLE/MAKEUP_1 + makeup fill logic in R203.

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: MAKEUP_TABLE.Enabled = true

Step	Action	Completion criteria
1	Accumulate ingots to RequiredFeet/RequiredRegular per MAKEUP_TABLE.Option (1: 11 feet/3 reg; 2: 4 feet/10 reg).	Makeup quantities reached

Required quantities per option are constants in R301 — confirm whether product-configurable.

6.8 Gripper & Pneumatics

4 gripper solenoids (extend/retract with reed-switch feedback) + laser air-blast, driven by the robot's gripper open/close ops via the SMC EX260 valve manifold. Source: R003/R008 + FB_D_Solenoid.

Control module	Class	Description
SMC Valve Manifold	valve_manifold	SMC EX260-SEN1 EtherNet/IP valve manifold — 5 solenoid points: grippers A–D + air-blast.
Gripper A Solenoid	double_solenoid	Gripper A extend/retract (FB_DO_SolenoidData[1]) with reed-switch feedback.
Laser Air-Blast Solenoid	single_solenoid	Laser lens air-blast (FB_DO_SolenoidData[5]).

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: TOP.CTRLAutomatic_Request = true

Step	Action	Completion criteria
1	Grippers extend/retract following KUKA_ROBOT.In.GRIPPER_x_OPN/CLS_OP; feedback echoed back as reed-switch status.	Gripper feedback matches command

6.9 ConveyTech Interface

6-step handshake (via robot Task 6) releasing a completed stack to the downstream ConveyTech conveyor / strapper. Mediated through the robot (KUKA_ROBOT.In.CONVEYOR_*). Source: R305.

State	Pattern	Notes
Idle	static	—
Execute	sequential	—
Aborted	static	Safe state — forced on safety-gate violation

EXECUTE SEQUENCE

Permissives: Enable_CONVE_INT = true AND Healthy_CONVE_INT = false

Step	Action	Completion criteria
1	Step 0→1 on Triger_CONVE_INT (robot CONVEYOR_INT or StackOffConeyor).	Step 1
2	Step 1→2: request robot Task 6 (MOV 6 → Task_Request).	Robot CONVEYOR_SIGNAL_REQ
3	Step 2→3→4: assert Output_CONVE_INT to the downstream conveyor.	Output acknowledged
4	Step 5→6→31→0: drop output, complete handshake, return to idle.	Handshake complete

SECTION 7

7 Alarms & interlocks

Trigger	Severity	Description	Action / response
K_RIO1.BNI006A_ConnectionFaulted	FAULT	K_RIO1 (IO-Link master 1) connection faulted	Latch until Fault_Reset_Request; investigate EtherNet/IP link.
K_RIO2.BNI006A_ConnectionFaulted	FAULT	K_RIO2 connection faulted	Latch until reset.
KUKA_ROBOT.ConnectionFaulted	FAULT	KUKA robot connection faulted	Cell stops; check robot EtherNet/IP.
K_VALVE_MANIFOLD.SOLV_Status	FAULT	Valve manifold status fault	Latch until reset.
SickAFX_Enc1_xyz:I1.ConnectionFaulted	FAULT	SICK encoder connection faulted	Latch until reset.
KUKA_ROBOT.In.ALARM_STOP	FAULT	Robot ALARM_STOP	Robot alarm; clear at robot + reset.
K_VALVE_MANIFOLD.SOLV_Undervoltage_Error	FAULT	Valve manifold under-voltage	Check 24V supply to manifold.
FB_DI_PushButtonData[1].OutPushButton_On	CRITICAL	E-Stop 1 (pushbutton) — MAJOR	Major fault; all EMs forced to aborted. Reset e-stop then Fault_Reset.
FB_DI_SensorData[23].OutSignal	CRITICAL	E-Stop 1 (safety input) — MAJOR	Major fault; all EMs forced to aborted.

SECTION 8

8 Approval & sign-off

Role	Name	Signature	Date
Prepared by	Pac Technologies		
Reviewed by			
Approved by (client)			

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